

Answer key

Version A

Marking guide and worked solutions · 50 points · **instructor copy: do not distribute before the exam**

How to mark this exam. Stages 1, 6 and parts of 2, 3, 4 and 5 assess reasoning, so there is no single right answer: give the mark when the choice is sensible **and defended with a reason or a number**. Apply **follow-through** from earlier answers. Accept rounding differences of ± 0.02 in r , ± 0.2 in t and ± 0.5 in forecasts. The reference sheet fixes the exam's rules of thumb ($|z| > 2$ unusual, $|t| > 2$ significant, $|r| > 0.7$ heavy overlap between predictors); the course may use others, so change them in the key and on the sheet together. "Objective" gives the syllabus learning outcome (LO) and your category: **Con** concept, **Cal** calculation, **Com** computation in Sheets.

1 · Quick answer sheet

Q	Objective	Expected answer	Pts	How the marks split
1	LO1, LO5 · Con	Four variables, each with a sign and a mechanism. Plausible set: temperature (+, hot days want cold treats), weekend (+, more leisure time), beach_event (+, crowds nearby), price (–, law of demand). Also fine: sunshine (+), rainfall (–), humidity (–), post_change (either, with a reason).	4	1 per row: sign and a coherent mechanism. A "wrong" sign with a good reason scores; a sign with no reason does not.
2	LO2, LO5 · Con	Best answers: iced_coffee_units (known only at closing time; caused by the same weather, so a symptom, not a cause) and gelato_revenue_eur (price $\times$ units contains the outcome: circular). Also accept post_change (just a date flag that overlaps with price) or a weak-mechanism variable (wind) with a reason from how the café works.	2	1 per variable: must give a logical reason. "Not correlated" earns 0 (no data seen yet).
3	LO1 · Con	A result that contradicts the stated expectation: a coefficient with the opposite sign and $ t > 2$, or a coefficient whose $ t < 2$ (not distinguishable from zero).	1	1: names a coefficient/t-statistic outcome, not a vague "if it does not work".
4	LO2 · Con	(a) Validity: the 0/1 coding treats a small yoga class and a 5,000-person festival the same, so it does not capture crowd size. (b) Reliability: two staff may disagree on whether a small gathering counts as "organised"; fix by recording expected attendance.	2	1 validity (construct not captured) + 1 reliability (inconsistent recording).
5	LO3 · Cal	(a) Rainfall $z = (14.0 - 0.76) \div 1.92 = 6.89$; temperature $z = (31.8 - 22.80) \div 3.43 = 2.62$. (b) Rainfall: mean 0.76 > median 0.00, right-skewed . Temperature: mean 22.80 $\approx$ median 22.90, roughly symmetric.	2	1 both z-scores + 1 shapes with the mean/median evidence.
6	LO3, LO4, LO5 · Con + Com	(a) 133 of 168 days (79%) are exactly 0, so rainfall does not separate most days; the distribution is strongly right-skewed (mean 0.76, median 0, max 14, $z = 6.9$); in Figure 1 the dots stack at 0 and only a few wet days sit to the right, so the slope rests on few points; $r^2 = 0.05$. (b) Almost all the evidence comes from the 11 storm days: without them the effect cannot be distinguished from zero ($ t < 2$), so the result is fragile. (c) =IF(E2>0,1,0). Accept equivalents such as =IF(E2>=0.1,1,0) or =--(E2>0).	3	1 explanation using Table 2/Figure 1 + 1 reading of the t-statistic change + 1 correct formula.
7	LO3, LO6 · Cal + Con	(a) $z = (294 - 191.44) \div 35.84 = 2.86$. (b) Keep it . The extreme is explained by its inputs (hot, weekend, event, highest price): Model 4 predicts 279, so the residual is only +15, about 0.8 SE; removing it barely moves the slope; deleting it would discard exactly the high-demand days the café has to plan for. Check the residual , not the raw value.	2	1 z + 1 reasoned decision using the facts.
8	LO3 · Com	(a) temperature_c, $r = 0.58$. (b) temperature_c and sunshine_hours ($r = 0.78$); gelato_price_eur and post_change ($r = 0.76$). Both pairs are needed.	2	1 (a) + 1 (b) both pairs.

9	LO3, LO5, LO6 · Con + Cal	(a) $r^2 = 0.78^2 = 0.61$: about 61% of the variation in sunshine moves with temperature; they largely measure the same thing (warm sunny days). (b) The model assumes predictors carry separate (independent) information. When two overlap, LINEST cannot tell which one is responsible: coefficients become unstable (they can shrink or flip sign), standard errors inflate and t-statistics fall even if the pair matters jointly. (c) Keep temperature (stronger r with gelato units: 0.58 vs 0.49; a direct mechanism for wanting a cold treat) and drop sunshine; keep gelato_price_eur (varies daily including the random price trials, the lever the owner controls) and drop post_change (a date flag).	3	1 + 1 + 1. (c) needs a choice for each pair with a reason.
10	LO3, LO5 · Cal	$b = 0.58 \times 35.84 \div 3.43 = 6.06$; $a = 191.44 - 6.06 \times 22.80 = 53.2$; $\hat{y}(28) = 53.2 + 6.06 \times 28 = 222.9$ (accept ± 1 ; LINEST gives 222.9).	2	1 slope + 1 intercept and prediction.
11	LO6 · Cal + Con	(a) $t(\text{temp}, M1) = 6.05 \div 0.66 = 9.2$; $t(\text{temp}, M2) = 5.28 \div 1.06 = 5.0$; $t(\text{sunshine}, M2) = 1.36 \div 1.47 = 0.9$. (b) Adjusted R^2 : $M1 = 1 - (1 - 0.335) \times 167 \div 166 = 0.331$; $M2 = 1 - (1 - 0.339) \times 167 \div 165 = 0.331$ (unchanged). (c) Sunshine adds almost no information beyond temperature (R^2 0.335 $\rightarrow$ 0.339) because the two overlap ($r = 0.78$). Temperature's SE rises from 0.66 to 1.06 and its t falls; sunshine itself is not significant. Drop sunshine.	3	1 three t-statistics + 1 two adjusted R^2 values + 1 explanation linking to the overlap.
12	LO5, LO6 · Com	(a) Row 1 is reversed: Col 1 = gelato_price_eur (−32.47), Col 2 = beach_event (29.83), Col 3 = weekend (33.47), Col 4 = temperature_c (7.12), Col 5 = intercept (151.66). (b) $t = -32.47 \div 4.47 = -7.3$: significant; holding temperature, weekend and event constant, each €1 higher price goes with about 32 fewer gelatos a day (a €0.30 change $\approx$ 10 units). (c) Any honest comparison with their own Stage 1 choices: supported expectations (temperature, weekend, event positive; price negative), or an expectation LINEST does not support, or an omitted variable that matters (price, event or weekend if left out).	3	1 all five matched + 1 t and interpretation "holding constant" + 1 look-back that names a specific expectation.
13	LO3, LO6, LO7 · Cal + Con	(a) $SE = \sqrt{(35.3^2/84 + 34.1^2/84)} = 5.36$; difference = 18.6; $t = 3.47$: significant. (b) The before/after comparison mixes the price rise with everything else that changed on 30 June: it was warmer (r between temperature and post_change = 0.56) and had more beach events ($r = 0.30$). Model 4 compares days with the same temperature, weekend and event, so it isolates price: demand rose with the summer despite the higher price. (c) Price is correlated with temperature ($r = 0.45$) and with the policy date: high-price days tend to be hot, high-demand days, so the positive confounding cancels the negative price effect and the simple $r \approx 0.02$. r near 0 does not mean "no effect" when another variable is tangled with it.	3	1 SE and t with the decision + 1 reconciliation (confounding by season/temperature) + 1 explanation of $r \approx 0$.
14	LO6 · Cal	May 21: predicted 182.8, residual +3.2, $ res < 39.3$: ordinary. Sep 4: predicted 183.5, residual −56.5, $ res > 39.3$: unusual . Action: investigate (stock-out, closure, forecast error, recording error); do not delete without a reason. About 1 day in 20 beyond 2 SE is expected even if the model is fine.	2	1 two predictions and residuals + 1 correct classification and sensible action.
15	LO4, LO6 · Con	(a) Add weekend first, with a positive sign: residual means +24.7 (weekend) vs −9.9 (weekday), a gap of 34.6, the largest; in Figure 2 the red weekend dots sit above 0 and the hollow weekday dots below. Model 1 under-predicts weekends and over-predicts weekdays. Next: beach_event (gap 25.5), then price (negative; gap 17.4). (b) Dots scattered randomly around the zero line, no pattern, weekend and weekday dots mixed, similar spread at every temperature.	2	1 variable + sign + evidence from the table or figure + 1 sketch showing a patternless cloud around 0.
16	LO5, LO6 · Cal	Baseline: Aug 3 244.4, Aug 8 195.7, Aug 18 210.5, Aug 19 203.1. $ errors $: baseline 39.4, 23.3, 70.5, 8.9 $\rightarrow$ MAE 35.5 ; M4 0.2, 16.5, 21.2, 20.0 $\rightarrow$ MAE 14.5 ; M5 3.6, 18.6, 21.5, 16.9 $\rightarrow$ MAE 15.2 .	5	1 baseline forecasts + 2 errors (≥ 10 of 12 correct = 2, 7–9 = 1) + 2 MAEs (all three = 2, two = 1).

17	LO6 · Con	(a) R^2 can only rise when variables are added: it measures fit to the days already seen, noise included, so it cannot show better prediction of new days. On the unseen days M5 is no better (15.2 vs 14.5 here; 14.4 vs 13.8 over all 56 unseen days). (b) Give the owner M4 : the same accuracy with half the inputs. It needs only the temperature forecast, the calendar and the café's own price; M5 needs four more weather forecasts that overlap with temperature, are themselves uncertain, and are harder to explain. The baseline (MAE 28.2 over all 56 days) is much worse than either model. A student who picks M5 must defend it with unseen-day evidence, which does not support it here.	2	1 in-sample vs unseen reasoning + 1 choice defended with accuracy and with what must be known in advance.
18	LO7 · Con	Plain-language memo: recommends temperature, weekend, beach_event and price (with why); gives a forecast error in cups (SE of $y \approx 20$ for M4, or $MAE \approx 14$ on unseen days); one limitation (a single April–September season of synthetic data; weather forecast error; association is not proof of cause; the model cannot foresee unusual days; do not extrapolate beyond the observed range).	3	1 recommendation with a reason + 1 a number in plain words + 1 a genuine limitation.
19	LO1, LO5, LO7 · Con	Open response. Strong answers name a plausible variable not in the table (e.g. forecast crowd size, school holidays, competitor price, cruise-ship arrivals, a heat-wave flag), say how it is measured, give a sign with a mechanism, propose adding it to LINEST (t-statistic, SE of y , adjusted R^2) and testing on unseen days (MAE), and say what result would count against it.	4	1 variable and measurement + 1 sign and mechanism + 1 testing plan + 1 falsifier.

2 · Worked tables

Q16, hold-out days (Table 8). Baseline = $0.5 \times (1 \text{ week ago}) + 0.3 \times (2 \text{ weeks ago}) + 0.2 \times (3 \text{ weeks ago})$.

Day	Actual	Baseline calculation	Baseline	M4	M5	errors (base / M4 / M5)
Aug 3	205	$0.5 \times 275 + 0.3 \times 201 + 0.2 \times 233$	244.4	204.8	208.6	39.4 / 0.2 / 3.6
Aug 8	219	$0.5 \times 166 + 0.3 \times 243 + 0.2 \times 199$	195.7	202.5	200.4	23.3 / 16.5 / 18.6
Aug 18	140	$0.5 \times 224 + 0.3 \times 201 + 0.2 \times 191$	210.5	161.2	161.5	70.5 / 21.2 / 21.5
Aug 19	212	$0.5 \times 195 + 0.3 \times 188 + 0.2 \times 246$	203.1	192.0	195.1	8.9 / 20.0 / 16.9
MAE						35.5 / 14.5 / 15.2

Q14, predictions from Model 4.

Day	Calculation	Predicted	Residual
May 21	$151.66 + (7.12 \times 24) + (33.47 \times 0) + (29.83 \times 0) + (-32.47 \times 4.3)$	182.8	+3.2
Sep 4	$151.66 + (7.12 \times 25) + (33.47 \times 0) + (29.83 \times 0) + (-32.47 \times 4.5)$	183.5	-56.5

3 · Where students go wrong

- **Reading LINEST Row 1 in the order of the X columns (Q12).** It is reversed, so the first number is the last X column. Give no mark for part (a) if the order is not reversed, but apply follow-through to (b).
- **Treating $|r| < 0.7$ as “no overlap” and ignoring negative correlations (Q8).** A pair at -0.75 counts.
- **“Delete the outlier” (Q7).** A raw extreme is not an error: ask whether the model explains it. Mark the reasoning, not the decision.
- **Dropping price because $r \approx 0$ (Q13c, Q1) or trusting the simple price slope (Q13b):** both ignore confounding.
- **“M5 is best because its R^2 is higher” (Q17).** Look for the argument that R^2 only rises with more variables and that what matters is unseen days.
- **Hypothesis marks:** never penalise a hypothesis because the data disagree; penalise a hypothesis that no data could disagree with.